③ Find the image of the region $y>1$ in the z-plane under the transformation $w=(1-i)z$.

Sol: Given $w=(1-i)z$

i.e., $u+iv=(x+iy)(1-i)$

$\Rightarrow u=x+y$ and $v=y-x$

or, $u+v=2y \Rightarrow y=\frac{u+v}{2}$

and $u-v=2x \Rightarrow x=\frac{u-v}{2}$.

Hence, the image of the region $y>1$ is $\frac{u+v}{2}>1$

i.e., $u+v>2$ in the w-plane.

3. Inversion:

The transformation $w=\frac{1}{z}$ represents the inversion.

Let $w=Re^{i\phi}$ and $z=re^{i\theta}$.

Then the transformation becomes

$$Re^{i\phi}=\frac{1}{r}e^{-i\theta}$$

so that $R=\frac{1}{r}$ and $\phi=-\theta$.

Thus under the transformation $w=\frac{1}{z}$, any point $P(r,\theta)$ in z-plane is mapped into the point $P'\left(\frac{1}{r},-\theta\right)$

Note that, the origin $z=0$ is mapped to the point $w=\infty$, called the point at infinity.

Examples

(1) Find the image of the circle $|z-1|=1$ under the transformation $w=\frac{1}{z}$.

Sol: Given $|z-1|=1$ is a circle centred at $(1,0)$ with radius 1.

~~Now~~ And, $w=\frac{1}{z} \Rightarrow z=\frac{1}{w}$

Now, $|z-1|=1 \Rightarrow \left|\frac{1}{w}-1\right|=1$

$$\Rightarrow |1-w|=|w|$$

$$\Rightarrow |1-(u+iv)|=|u+iv|$$

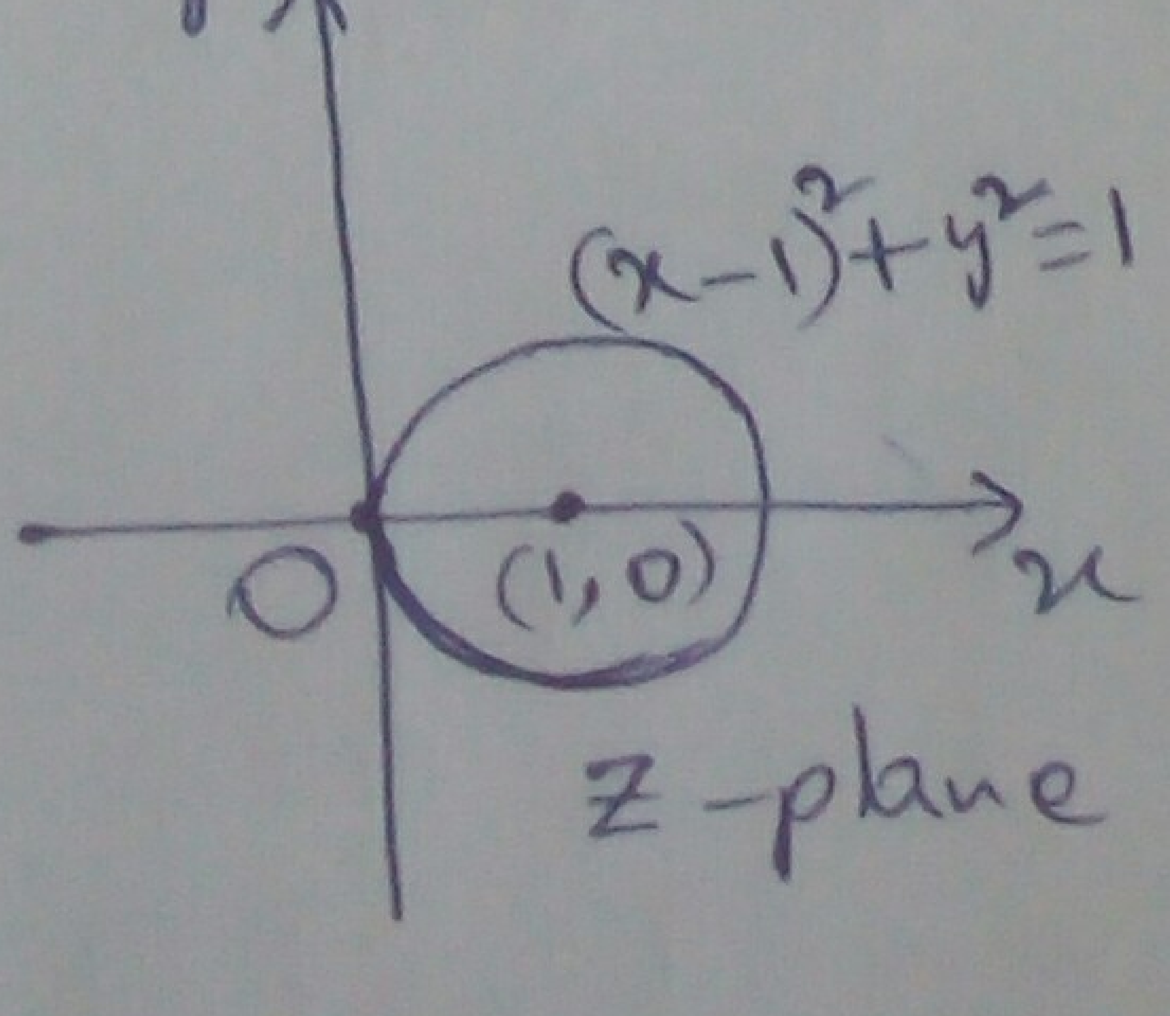
$$\Rightarrow \sqrt{(1-u)^2+(-v)^2}=\sqrt{u^2+v^2}$$

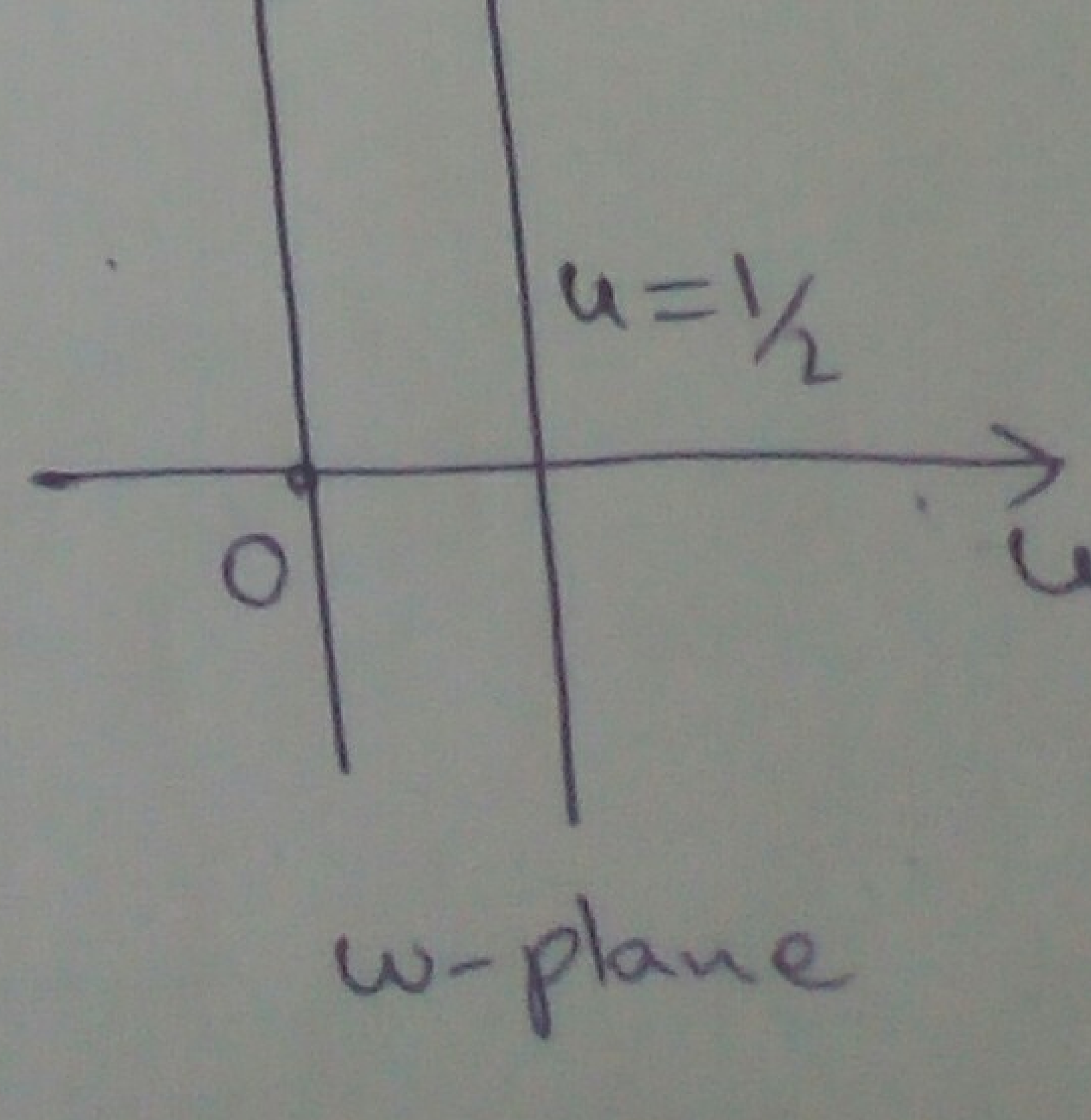
$$\Rightarrow (1-u)^2+v^2=u^2+v^2$$

$$\Rightarrow 2u=1$$

$$\Rightarrow u=\frac{1}{2}$$

which is a straight line in the w-plane





(2) Find the image of the infinite strip $\frac{1}{4} \le y \le \frac{1}{2}$ in the z-plane under the transformation $w = \frac{1}{z}$.

Sol: Given $w = \frac{1}{z}$

i.e., $z = \frac{1}{w}$

$$\Rightarrow x + iy = \frac{1}{u+iv}$$

$$\Rightarrow x + iy = \frac{u-iv}{u^2+v^2}$$

$$\Rightarrow x = \frac{u}{u^2+v^2} \text{ and } y = \frac{-v}{u^2+v^2}$$

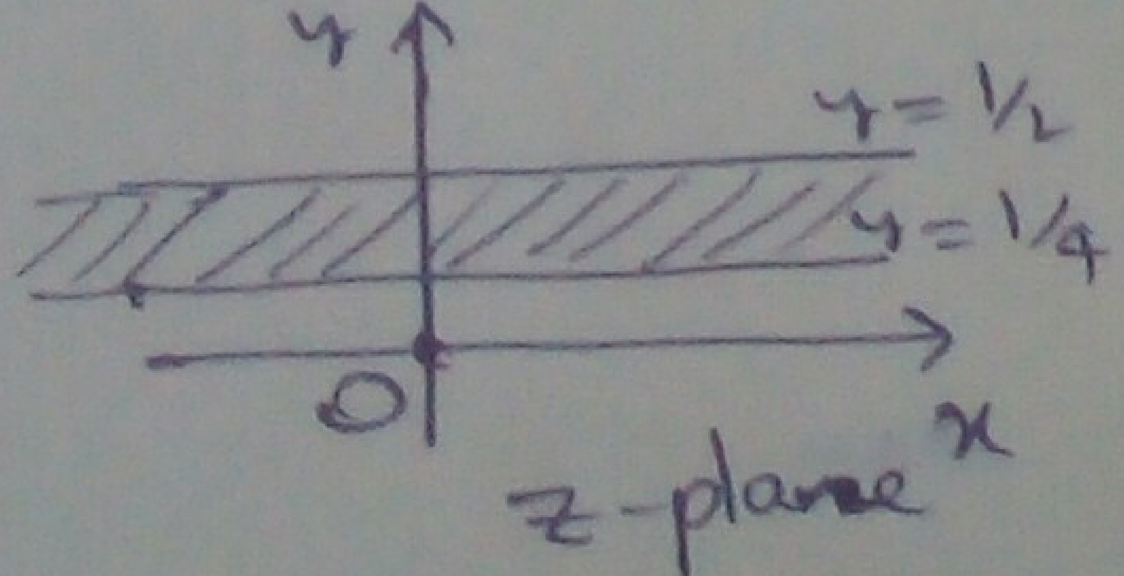
when $y = \frac{1}{4}$, $\frac{-v}{u^2+v^2} = \frac{1}{4} \Rightarrow u^2 + (v+2)^2 = 4$

which is a circle centre is at $(0,-2)$ and radius is 2 units in the w-plane

when $y = \frac{1}{2}$, $\frac{-v}{u^2+v^2} = \frac{1}{2} \Rightarrow u^2 + (v+1)^2 = 1$

which is a circle centred at $(0,-1)$ and radius is 1 unit in the w-plane.

Hence, the infinite strip $\frac{1}{4} < y < \frac{1}{2}$ is transformed into the region in between the circles $u^2 + (v+2)^2 = 4$ and $u^2 + (v+1)^2 = 1$ in the w-plane.



4. Exponential:

The transformation $w=e^z$ represents the exponential.

Examples:

① Find the image of the region in the z-plane between the lines $y=0$ and $y=\frac{\pi}{2}$ under the transformation $w=e^z$.

Sol: Given $w=e^z$

$\Rightarrow Re^{i\phi}=e^{x+iy}$

$\Rightarrow Re^{i\phi}=e^{x}.e^{iy}$

$w=e^z$

$\Rightarrow u+iv=e^{x+iy}$

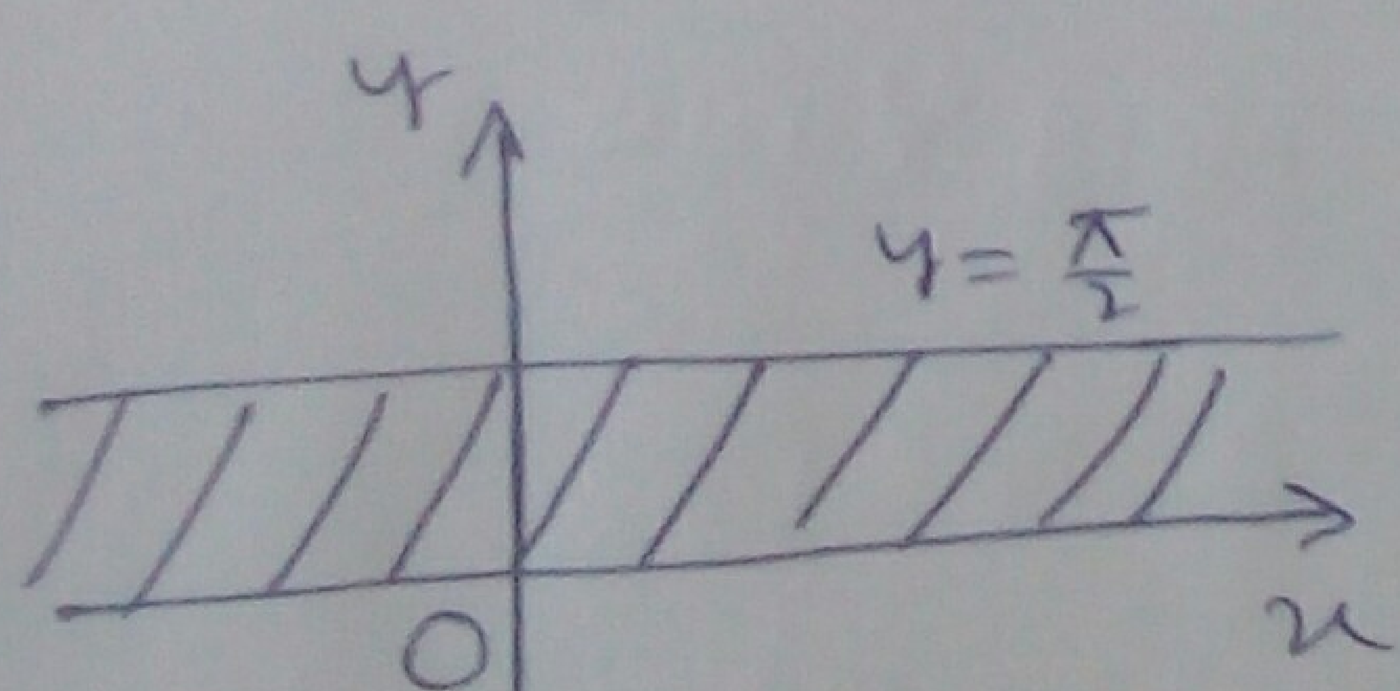
$\Rightarrow u+iv=e^{x}(\cos y+i\sin y)$

$\Rightarrow u=e^{x}\cos y$ and

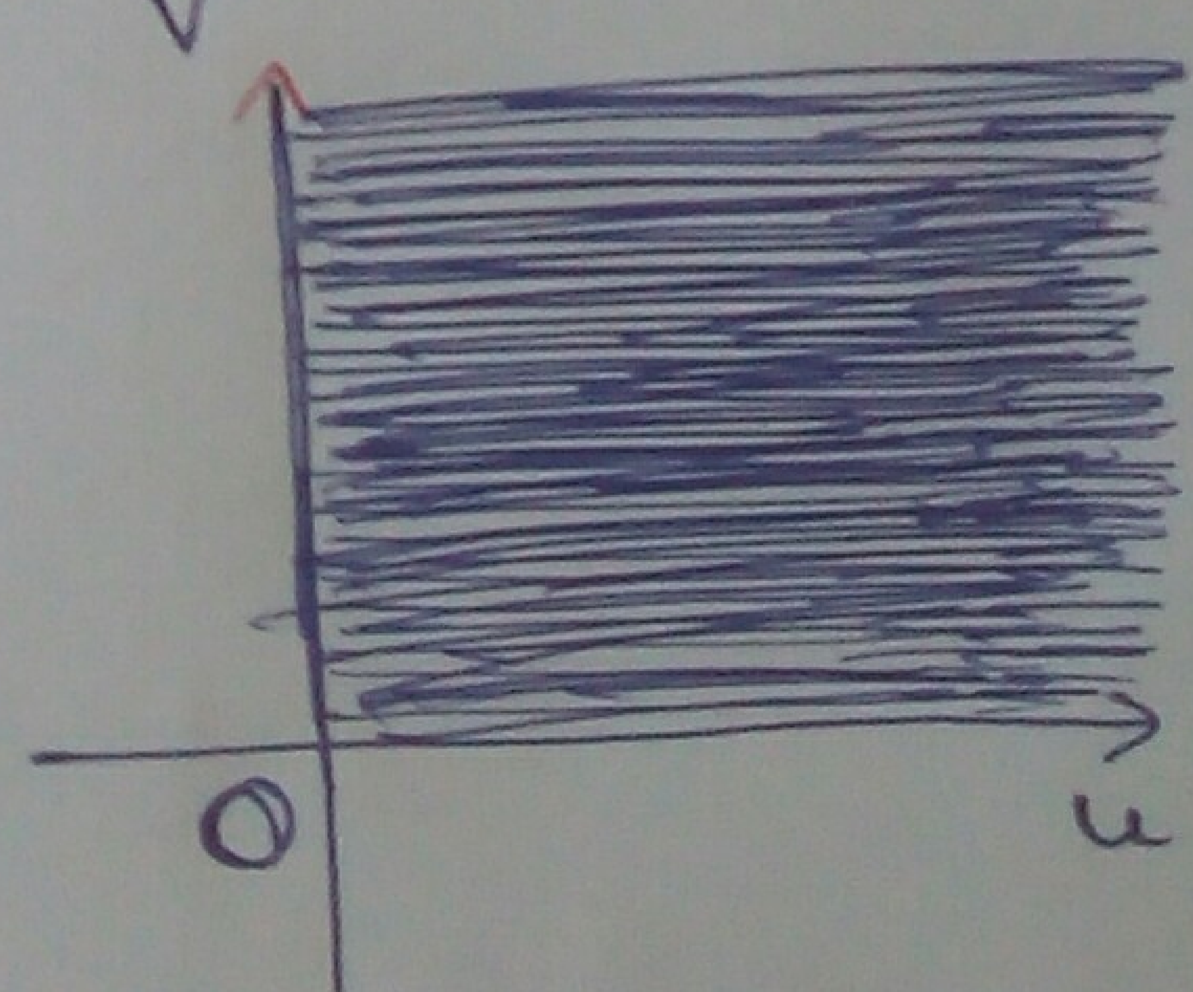
$v=e^{x}\sin y$.

which implies that $R=e^{x}$ and $\phi=y$.

$y=0 \Rightarrow \phi=0$ and $y=\frac{\pi}{2}\Rightarrow\phi=\frac{\pi}{2}$



z-plane



w-plane

Therefore, the infinite strip between the lines $y=0$ and $y=\frac{\pi}{2}$ is mapped into the first quadrant of w-plane.

(2) Find the image of the infinite strip between the lines $x=1$ and $x=3$ under the transformation $w=z^2$.

Sol: Given $w=z^2$.

i.e., $u+iv=(x+iy)^2 \Rightarrow u+iv = x^2-y^2+i2xy$

$\Rightarrow u = x^2-y^2$ and $v=2xy$

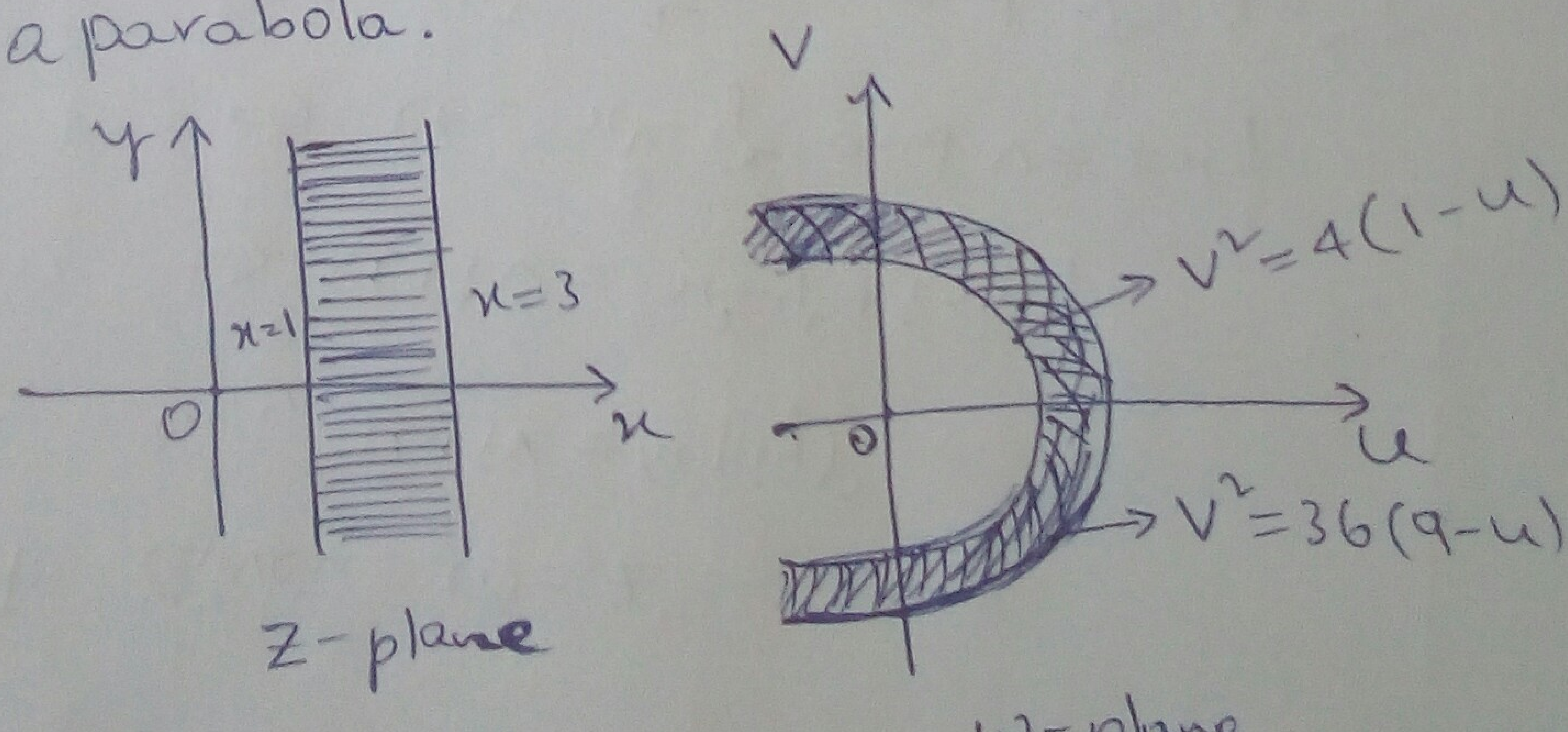
$\Rightarrow y^2 = x^2-u$ and $v^2=4x^2y^2$

So,

$v^2=4x^2y^2 \Rightarrow v^2 = 4x^2(x^2-u)$.

Now, $x=1 \Rightarrow v^2 = 4(1-u)$ which is a parabola whose vertex is $(1,0)$, the focus is $(0,0)$ and the latus rectum is 4.

and, $x=3 \Rightarrow v^2 = 36(9-u)$ which is also a parabola.



Therefore, the infinite strip between the lines $x=1$ and $x=3$ is mapped into the region between the parabolas $v^2=4(1-u)$ and $v^2=36(9-u)$.